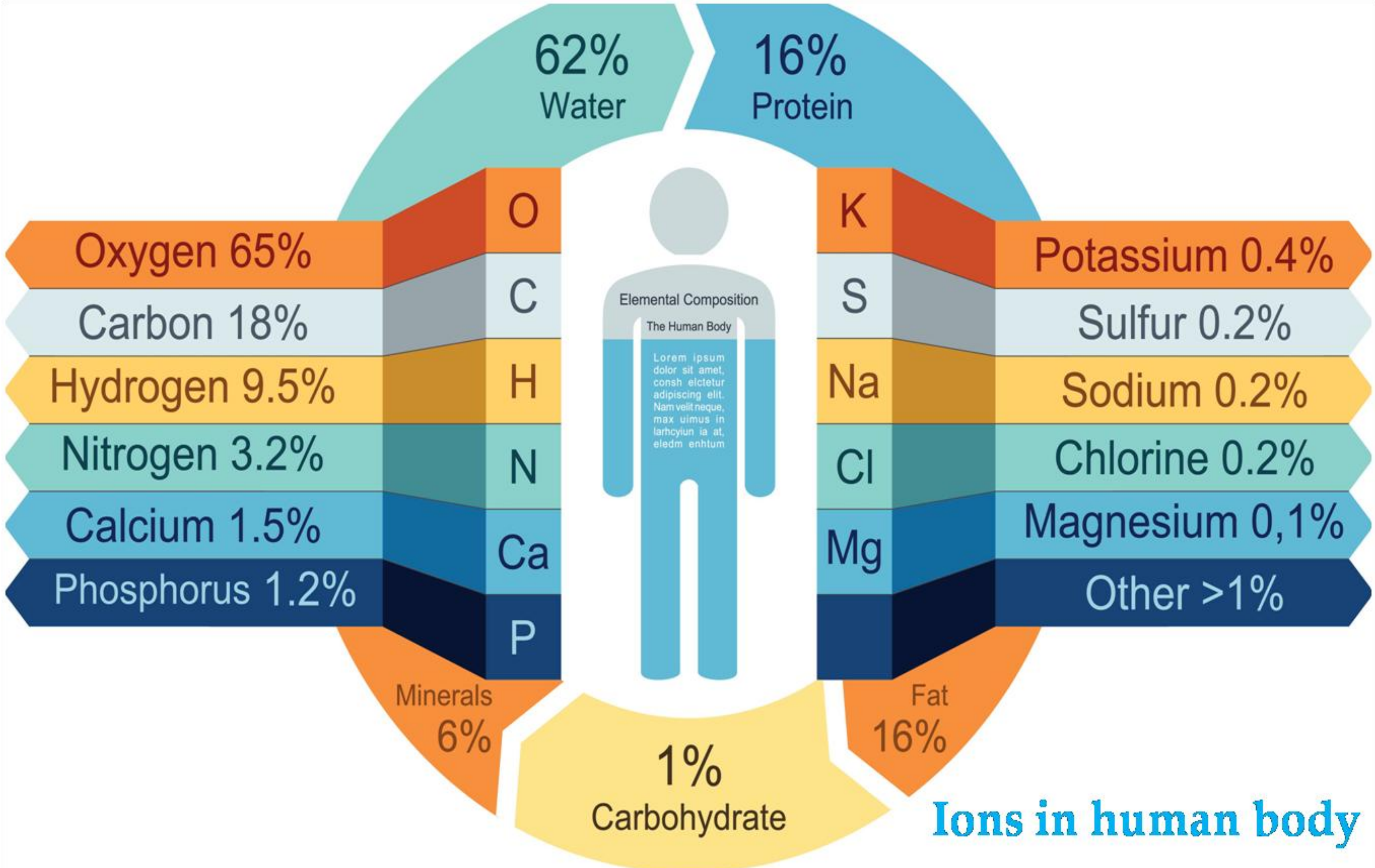
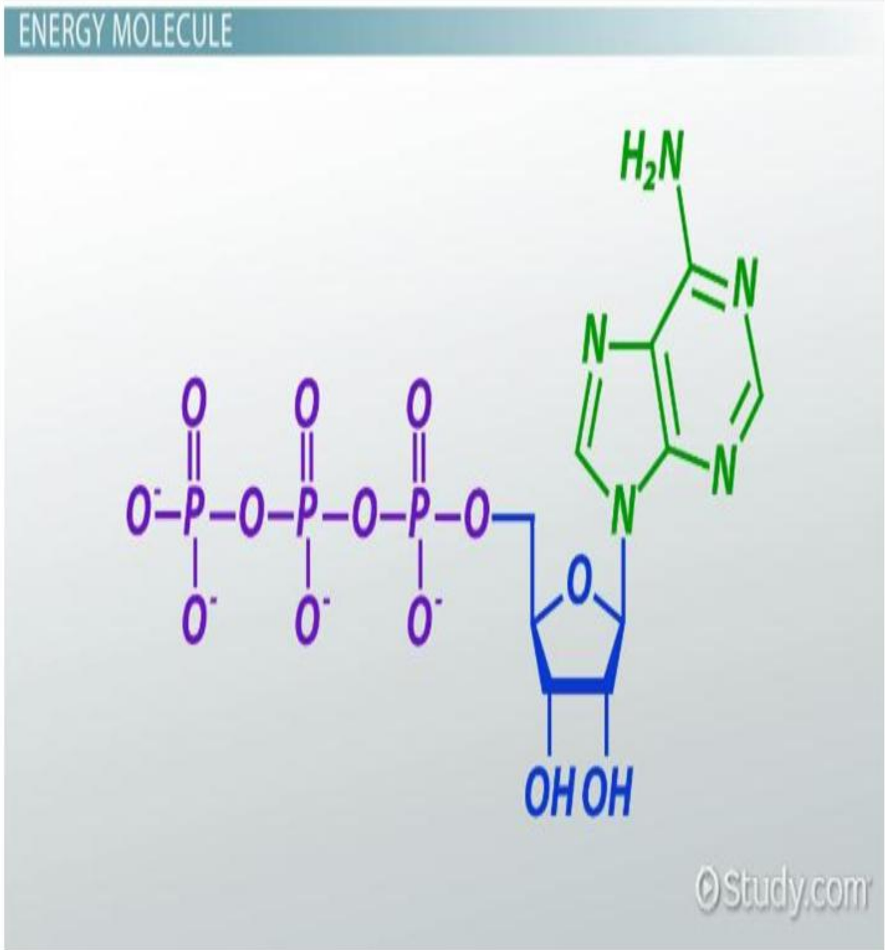
MINERAL IONS



Outline the role of minerals and ions in biological systems.

1. They are ***components of smaller molecules*** e.g. phosphorus is contained in ATP and iodine is contained in thyroxin, etc.



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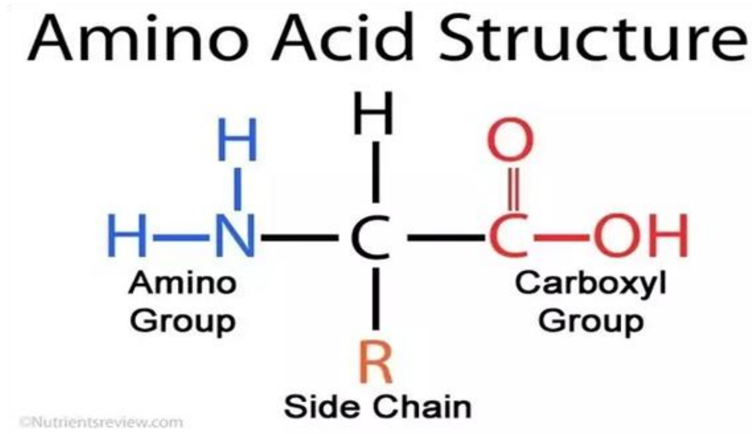
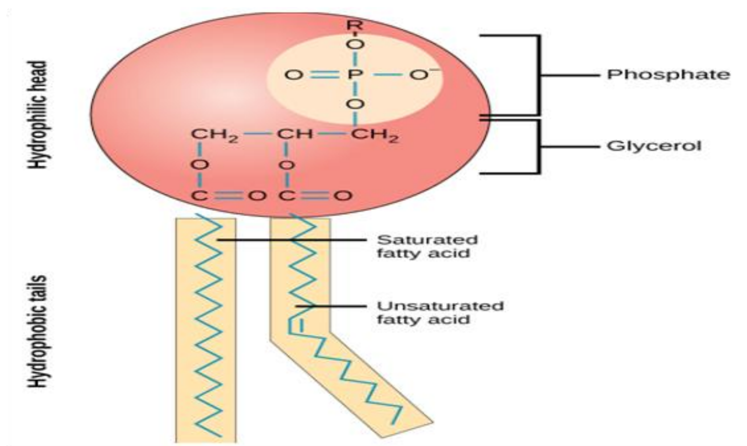
2. They are constituents of **large molecules** e.g. proteins contain nitrogen and sulphur, phospholipids contain phosphorus, nucleic acids contain nitrogen and phosphorus, etc.





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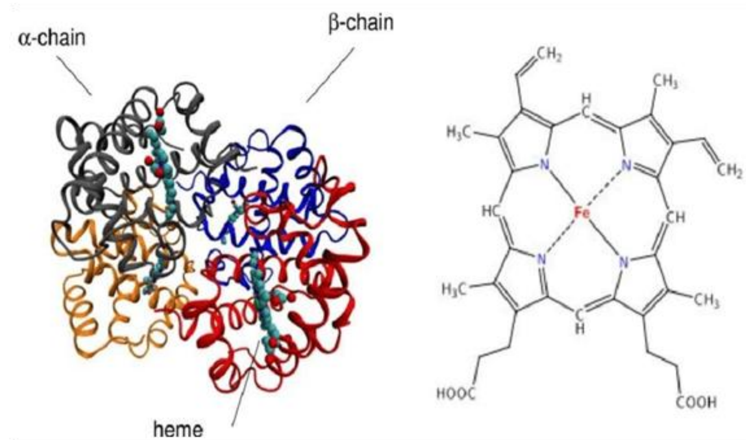
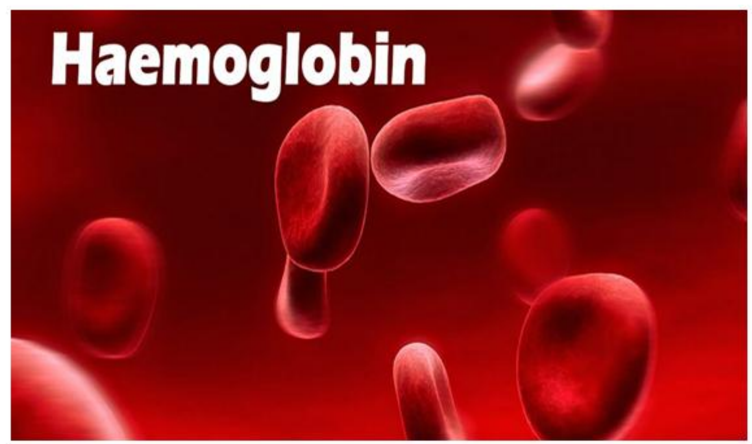
3. They are components of pigments e.g. ***haemoglobin*** and cytochromes which contain iron, chlorophyll contain magnesium, etc.





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4. *They are **metabolic activators** e.g. ATP activates glucose before it is broken down in cell respiration, calcium ions activate ATPase enzyme during muscle contraction.*

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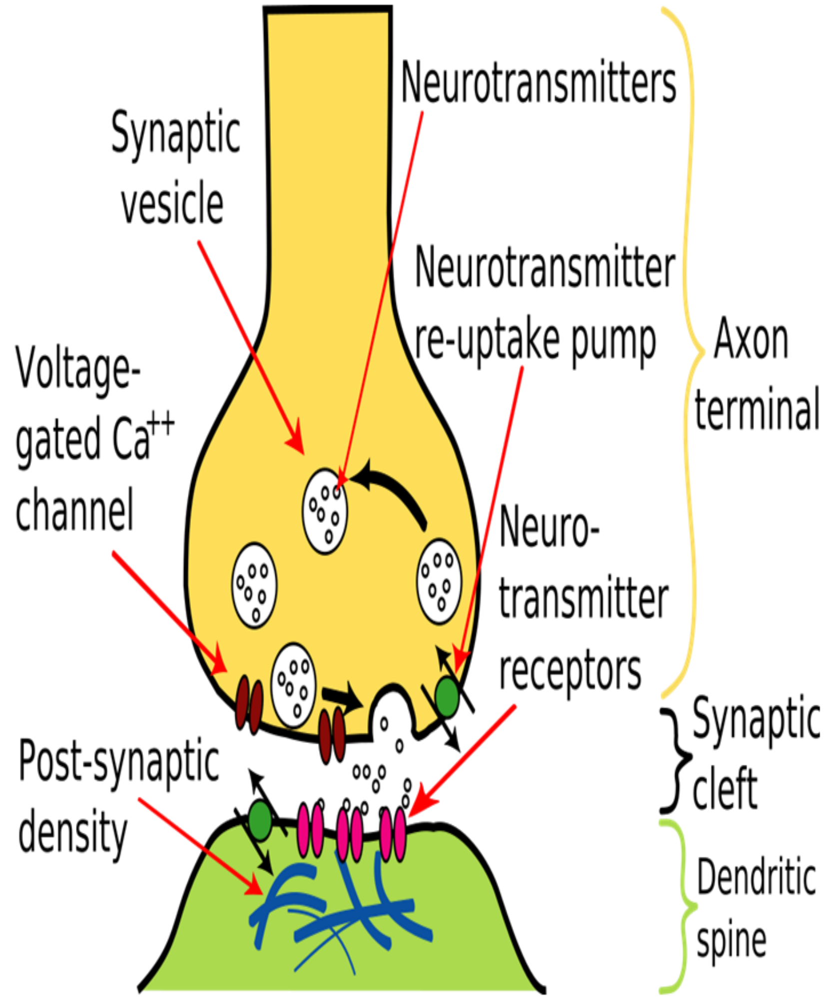
5. *They determine the **anion, cation balance** e.g. Na^{+}, K^{+} and Ca^{+} are important in transmission of impulses and muscle contraction.*



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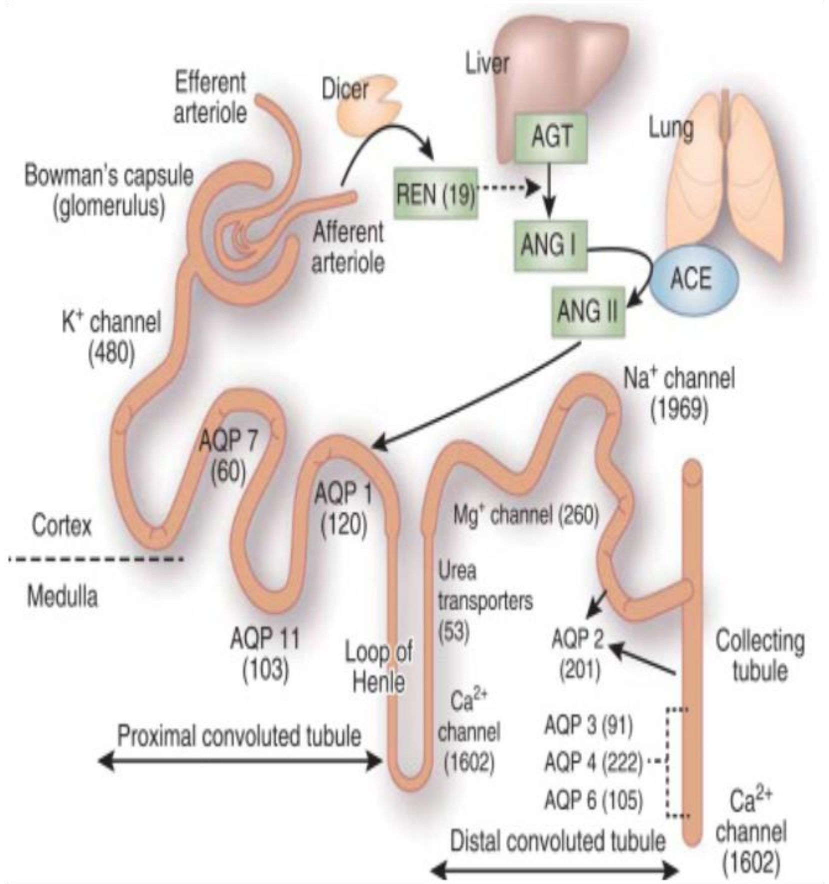
6. They determine the **osmotic pressure** and water potential so that it does not fluctuate beyond narrow limits e.g. Na^+, K^+ and Cl^- are involved in water balance in the kidneys.



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7. They are constituents of structures **in cell membranes,** cell walls, bones, enamel and shells.